VI SEMESTER (OPEN ELECTIVE-II)	L	T	P	C
	2	-	2	3
20ME6O01 - BASICS OF 3D PRINTING				

COURSEOBJECTIVES:

1. To explore technology used in additive manufacturing.
2. ToacquireknowledgeforselectingcorrectCADformatsinmanufacturingprocess.
3. To understand the operating principles and limitations of liquid, solid and laser based additive manufacturing system.
4. To design the process of additive manufacturing including tools used for design.
5. Toacquireknowledgeonimportantprocessparametersforbio-manufacturing

COURSEOUTCOMES:

After successful completion of this course, students should be able to:

CO1: To impart the fundamentals of Additive Manufacturing Technologies for engineering applications [K2].
CO2: Select and use correct CAD for mats in the manufacture of a3D printed part [K2].
CO3: Explain the operating principles, capabilities, and limitations of liquid, solid and laser based additive manufacturing system[K2].
CO4: Enumerate the design process for additive manufacturing including tools used for design and some features required for design [K2].
CO5: Describe the important process parameters for bio-manufacturing and determine the suitable additive technique for bio-manufacturing, aerospace and manufacturing engineering[K2].

UNIT I
INTRODUCTION

3D printing Overview, History, Need, Classification, Additive Manufacturing Technology in product development, Materials for Additive Manufacturing Technology.

UNIT II
REVERSE ENGINEERING

BasicConcept–3DScanning Digitization techniques, Model Reconstruction, Data Processing for Additive Manufacturing Technology, Part Orientation and support generation, Model Slicing, Tool path Generation.

UNIT III
ADDITIVE MANUFACTURING SYSTEMS

SOLID& LIQUID BASED- Classification, Stereo lithography Apparatus (SLA) - Principle, process, advantages, Fused Deposition Modeling –Principle, process, advantages.
LASERBASED- Selective Laser Sintering–Principle, Process, advantages, Three Dimensional Printing –Principle, process, advantages – Laser Engineered Net Shaping (LENS).

UNIT IV
DESIGN FOR AM

Motivation, Design for Manufacturing and Assembly (DFMA)-concepts and objectives, AM unique capabilities, Exploring design freedoms, Design tools for AM-Part Orientation, Removal of Supports, Hollowing out parts, Inclusion of Undercuts, Other Manufacturing Constraining Features, Interlocking Features, Reduction of Part Count in an Assembly, Identification of markings/numbers etc.

UNIT V
APPLICATIONS OF 3D PRINTING

Customized implants and prosthesis: Design and development, Bio-Additive Manufacturing- Computer Aided Tissue Engineering (CATE), Applications of 3D Printing in Aerospace, Automotive, Manufacturing and Architectural Engineering.

Text Books:

1. Patri K. Venuvinod., and Weiy in Ma., Rapid prototyping Laser based and other Technologies, First Edition, Springer Science +Business Media, LLC, 2004.
2. Chua C.K., Leong K.F., and Lim C.S., Rapid prototyping: Principles and applications, Third Edition, World Scientific Publishers, 2016.
3. Gebhardt A, Rapid prototyping, Hanser Gardener Publications,2017.
4. Chee Kai Chua, Kah Fai Leong, 3D Printing and Additive Manufacturing: Principles and Applications, World Scientific Publishers, Fourth Edition of Rapid Prototyping, 2018.

Reference Books:

1. LiouL.W.and LiouF.W.,Rapid Prototyping and Engineering applications:A toolbox for prototype development, CRC Press, 2017.
2. Kamrani A.K. and Nasr E.A., Rapid Prototyping: Theory and practice, Springer, 2016.
3. Hilton P.D. and Jacobs P.F., Rapid Tooling: Technologies and Industrial Applications, CRC press, 2015.

Web References:

1. https://all3dp.com/
2. https://www.thingiverse.com/
3. https://additivemanufacturing.com/